

Titanic Dataset Analysis

Step 1: Load & Preview the Data

- Loaded the dataset successfully with **891 rows** and **12 columns**.
- Columns include info like: Survived, Pclass, Name, Sex, Age, Fare, etc.
- Got a quick look at the first few rows.

Step 2: Data Cleaning (next step)

We will:

- Identify and count **missing values**.
- Decide how to handle them (e.g., fill, drop, or flag).
- Check and possibly adjust **data types**.

Step 3: Exploratory Data Analysis (EDA)

We'll explore:

- Overall survival rate.
- Survival by **gender**, **class**, **age**, etc.
- Visual insights: bar plots, histograms, pie charts.

Step 4: Feature Engineering

We'll create new features like:

- Title from the Name (Mr., Mrs., Miss, etc.)
- Family Size (SibSp + Parch + 1)
- Convert text (like Sex and Embarked) into numeric codes.

Step 5: Correlation & Insights

We'll look at:

- How different features relate to survival.
- Which features are the most important predictors.

Step 6 (Optional): Model Building

If you're up for it, we can build a simple machine learning model (e.g., logistic regression or decision tree) to **predict survival**.